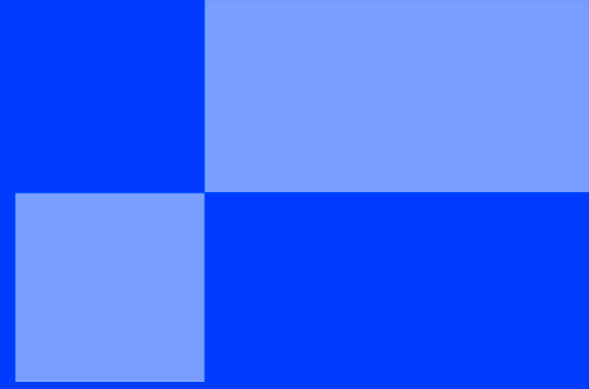
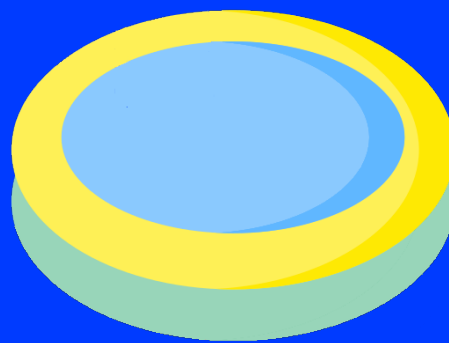




PREPARATION BOOTCAMP

< SECOND PROGRAM />



PREPARATION BOOTCAMP

GAME



Hack The Box

In addition to the tasks below, we encourage you to discover the [Root Me Platform](#) which proposes some cybersecurity challenges. Try to go as far as possible!

Work on it as soon as you have a bit of time, or whenever you need a break in your day!



Your objective is to improve your hangman game. You should:

- ✓ reuse your previously written program or rewrite it from scratch if necessary.
- ✓ check out pygame's official [documentation](#) and [tutorials](#).

Random word



Add a filename as a command-line argument to your program.
The program should read the words from this file and randomly select one for guessing.



Each line of the file should contain one and only one word, but obviously, you should handle any unexpected file format (or missing file or whatever).

```
$> python3 hangman.py my_wordlist.txt
...
$> python3 hangman.py my_other_wordlist.txt
...
$> python3 hangman.py
Error: missing argument
$>
```



Standard error output...

High score



When ending a game, if the high score is beaten : add a new line into an external file, with at least the number of attempts and the date. Then, write on the standard output:

Best ever! You guessed '<WORD>' in <X> attempts..

If the high score is not beaten, write:

You guessed '<WORD>' in <X> attempts, but the record from <YYYY-MM-DD> is <Y> attempts .



CHALLENGE



This challenge is focused on error handling and program security. If you don't feel up to it straight away, that's normal. Come back later to try again. You should compare your productions with others and discuss the pros and cons of each solution. The best answers could be presented publicly.

- ✓ Try to break your program or make it crash.
- ✓ Like any user could, do some crazy dumb stuff with the program.
- ✓ Test invalid inputs, special characters, wrong types and arguments, etc.
- ✓ Try to abuse and exploit your program in unintended ways (privileges, access, ...).
- ✓ Then, make your program unbreakable, secure and solid as a rock.



You can also ask someone around you to try and get a bug from your program. This is beta testing, and this phase is necessary before delivering a program.



Full game



Now that you have covered the basics, reuse what you've made before and use what you have learned in order to implement a graphical version of your hangman game. You can design it freely, and get inspiration from the Internet.



Package your application



Turn your pygame program into a Windows executable and/or a Linux binary file.

Go further



Customize your app by adding some nice and cool features, such as:

- ✓ a menu,
- ✓ a timer,
- ✓ a time limit,
- ✓ a live score,
- ✓ a life bar,
- ✓ a leaderboard with the best scores,
- ✓ a difficulty level,
- ✓ a basic AI engine to help you solve the game,
- ✓ anything fun and/or valuable.

v 2.2

{EPITECH}